

EE4402 Tutorial One Solutions

Q1. Let $\hat{x}(t)$ is the Hilbert transform of a real signal $x(t)$. Prove the results below.

- a) If $x(t)$ is even symmetric, $\hat{x}(t)$ is odd symmetric. If $x(t)$ is odd symmetric, $\hat{x}(t)$ is even symmetric.

Since $\hat{x}(t) = x(t) * \frac{1}{\pi t}$, if $x(-t) = x(t)$, $\hat{x}(-t) = x(-t) * \frac{1}{\pi(-t)} = -x(t) * \frac{1}{\pi t} = -\hat{x}(t)$, which shows $\hat{x}(t)$ is odd symmetric.

The same method can be used to show that if $x(t)$ is odd symmetric, $\hat{x}(t)$ is even symmetric.

- b) If $x(t) = \cos(2\pi f_0 t)$ for positive f_0 , then $\hat{x}(t) = \sin(2\pi f_0 t)$.

If $x(t) = \cos(2\pi f_0 t)$, $X(f) = \frac{1}{2}[\delta(f - f_0) + \delta(f + f_0)]$ where $\delta(f - f_0)$ is a Dirac delta function at positive frequency f_0 and $\delta(f + f_0)$ is a Dirac delta function at the (negative) frequency $-f_0$.

$$\begin{aligned}\hat{X}(f) &= -j \operatorname{sgn}(f) X(f) = \frac{1}{2}(-j \operatorname{sgn}(f) \delta(f - f_0) - j \operatorname{sgn}(f) \delta(f + f_0)) \\ &= \frac{1}{2}(-j \delta(f - f_0) + j \delta(f + f_0)) = \frac{1}{2} \left(e^{-\frac{j\pi}{2}} \delta(f - f_0) + e^{\frac{j\pi}{2}} \delta(f + f_0) \right)\end{aligned}$$

Since $\delta(f - f_0) \leftrightarrow e^{j2\pi f_0 t}$ and $\delta(f + f_0) \leftrightarrow e^{-j2\pi f_0 t}$,

$$\begin{aligned}\hat{x}(t) &= \frac{1}{2} \left(e^{-\frac{j\pi}{2}} e^{j2\pi f_0 t} + e^{\frac{j\pi}{2}} e^{-j2\pi f_0 t} \right) = \frac{1}{2} \left(e^{j(2\pi f_0 t - \frac{\pi}{2})} + e^{-j(2\pi f_0 t - \frac{\pi}{2})} \right) \\ &= \cos \left(2\pi f_0 t - \frac{\pi}{2} \right) = \sin(2\pi f_0 t)\end{aligned}$$

- c) $\hat{\hat{x}}(t) = -x(t)$.

Since $\hat{X}(f) = -j \operatorname{sgn}(f) X(f)$, $\hat{\hat{X}}(f) = -j \operatorname{sgn}(f) \hat{X}(f) = -\operatorname{sgn}^2(f) X(f) = -X(f)$ and $\hat{\hat{x}}(t) = -x(t)$.

d) $\int_{-\infty}^{\infty} x^2(t)dt = \int_{-\infty}^{\infty} \hat{x}^2(t)dt.$

By Rayleigh energy theorem,

$$\begin{aligned}\int_{-\infty}^{\infty} \hat{x}^2(t)dt &= \int_{-\infty}^{\infty} |\hat{X}(f)|^2 df = \int_{-\infty}^{\infty} \hat{X}(f)\hat{X}^*(f) df \\ &= \int_{-\infty}^{\infty} -j\text{sgn}(f)X(f) \times (j\text{sgn}(f)X^*(f))df = \int_{-\infty}^{\infty} \text{sgn}^2(f)X(f)X^*(f)df \\ &= \int_{-\infty}^{\infty} |X(f)|^2 df = \int_{-\infty}^{\infty} x^2(t)dt\end{aligned}$$

e) $x(t)$ and $\hat{x}(t)$ are orthogonal.

Let $x(t) \leftrightarrow X(f)$ and $\hat{x}(t) \leftrightarrow \hat{X}(f)$.

$$x(t) = \int_{-\infty}^{\infty} X(f_1)e^{j2\pi f_1 t} df_1 \text{ and } \hat{x}^*(t) = \int_{-\infty}^{\infty} \hat{X}^*(f)e^{-j2\pi f t} df.$$

$$\begin{aligned}\langle x(t), \hat{x}(t) \rangle &= \int_{-\infty}^{\infty} x(t)\hat{x}^*(t)dt \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(f_1)e^{j2\pi f_1 t} df_1 \int_{-\infty}^{\infty} \hat{X}^*(f)e^{-j2\pi f t} df dt \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(f_1)\hat{X}^*(f)e^{j2\pi f_1 t} e^{-j2\pi f t} dt df_1 df \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(f_1)\hat{X}^*(f) \left(\int_{-\infty}^{\infty} e^{j2\pi(f_1-f)t} dt \right) df_1 df \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(f_1)\hat{X}^*(f)\delta(f_1-f)df_1 df \\ &= \int_{-\infty}^{\infty} X(f)\hat{X}^*(f)df = \int_{-\infty}^{\infty} X(f)(j\text{sgn}(f)X^*(f))df \\ &= j \int_{-\infty}^{\infty} \text{sgn}(f)|X(f)|^2 df\end{aligned}$$

Since $x(t)$ is a real function, $X(f)$ is Hermitian, which means $|X(f)|^2$ is an even function of f . Because $\text{sgn}(f)$ is an odd function, $\text{sgn}(f)|X(f)|^2$ is an odd function of f . The

integration of an odd function over $(-\infty, \infty)$ is zero, ie., $\langle x(t), \hat{x}(t) \rangle = 0$ and they are orthogonal.

Q2. Assume T is a positive constant and $x_i \in \{0,1\}$ for $0 \leq i \leq 2$. [Hint: $\text{sinc}\left(\frac{t}{T}\right) \leftrightarrow T \times \Pi(fT)$]

a) Find energy of the signal $\text{sinc}(t)$.

From the hint, we have $\text{sinc}(t) \leftrightarrow \Pi(f)$. The energy of $\text{sinc}(t)$, by definition, is

$$\begin{aligned} E_1 &= \langle \text{sinc}(t), \text{sinc}(t) \rangle = \int_{-\infty}^{\infty} \text{sinc}(t) \text{sinc}^*(t) dt \quad (\text{sinc}(t) \text{ is even.}) \\ &= \int_{-\infty}^{\infty} \text{sinc}(\tau) \text{sinc}(t - \tau) d\tau \big|_{t=0} \\ &= \text{sinc}(t) * \text{sinc}(t) \big|_{t=0} = \mathcal{F}^{-1}[\Pi(f) \times \Pi(f)] \big|_{t=0} \\ &= \mathcal{F}^{-1}[\Pi(f)] \big|_{t=0} = \text{sinc}(t) \big|_{t=0} = 1. \end{aligned}$$

b) Show that $\text{sinc}\left(\frac{t}{T}\right)$ and $\text{sinc}\left(\frac{t-T}{T}\right)$ are orthogonal.

If $\langle \text{sinc}\left(\frac{t}{T}\right), \text{sinc}\left(\frac{t-T}{T}\right) \rangle = 0$, they will be orthogonal by definition.

$$\begin{aligned} \langle \text{sinc}\left(\frac{t}{T}\right), \text{sinc}\left(\frac{t-T}{T}\right) \rangle &= \int_{-\infty}^{\infty} \text{sinc}\left(\frac{t}{T}\right) \text{sinc}\left(\frac{t-T}{T}\right) dt \\ &= \int_{-\infty}^{\infty} \text{sinc}\left(\frac{\tau}{T}\right) \text{sinc}\left(\frac{T-\tau}{T}\right) d\tau = \text{sinc}\left(\frac{t}{T}\right) * \text{sinc}\left(\frac{t}{T}\right) \big|_{t=T} \end{aligned}$$

Since $\text{sinc}\left(\frac{t}{T}\right) \leftrightarrow T \times \Pi(fT)$.

$$\begin{aligned} \langle \text{sinc}\left(\frac{t}{T}\right), \text{sinc}\left(\frac{t-T}{T}\right) \rangle &= \mathcal{F}^{-1}[(T \times \Pi(fT)) \times (T \times \Pi(fT))] \big|_{t=T} \\ &= \mathcal{F}^{-1}[T^2 \times \Pi(fT)] \big|_{t=0} = T \times \text{sinc}\left(\frac{t}{T}\right) \big|_{t=T} \\ &= T \times \text{sinc}(1) = 0 \end{aligned}$$

c) Propose a set of orthonormal basis functions for signals $s(t) = \sum_{i=0}^2 x_i \times \text{sinc}\left(\frac{t-iT}{T}\right)$.

Find the size of the signal set $\{s(t)\}$ and its average energy assuming equiprobable signals.

From b), we have $\text{sinc}\left(\frac{t}{T}\right)$ and $\text{sinc}\left(\frac{t-T}{T}\right)$ are orthogonal. We also can show that $\text{sinc}\left(\frac{t}{T}\right)$, $\text{sinc}\left(\frac{t-T}{T}\right)$ and $\text{sinc}\left(\frac{t-2T}{T}\right)$ are all orthogonal. So we can choose them as orthogonal bases.

By similar argument as in a), we have $E_T = \langle \text{sinc}\left(\frac{t}{T}\right), \text{sinc}\left(\frac{t}{T}\right) \rangle = T$. It is obvious that $\text{sinc}\left(\frac{t}{T}\right)$, $\text{sinc}\left(\frac{t-T}{T}\right)$ and $\text{sinc}\left(\frac{t-2T}{T}\right)$ all have energy T because the latter two are time-shifting versions of $\text{sinc}\left(\frac{t}{T}\right)$. The basis functions can be

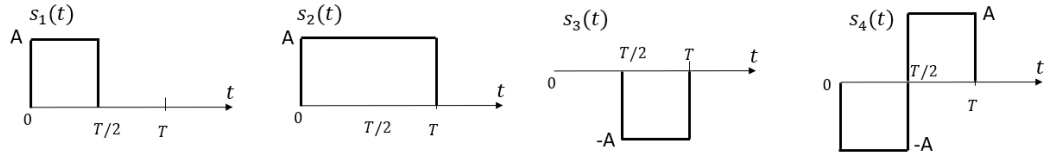
$$\Phi = \{\phi_1(t) = \sqrt{\frac{1}{T}} \text{sinc}\left(\frac{t}{T}\right), \phi_2(t) = \sqrt{\frac{1}{T}} \text{sinc}\left(\frac{t-T}{T}\right), \phi_3(t) = \sqrt{\frac{1}{T}} \text{sinc}\left(\frac{t-2T}{T}\right)\}$$

Since $x_i \in \{0,1\}$, and size of Φ is 3, we have the size of signal set is $2^3 = 8$.

Their vector forms are $[0,0,0], [1,0,0], [0,1,0], [1,1,0], [0,1,1], [1,1,1], [0,0,1]$ and $[1,0,1]$. Since all signals are equiprobable, the average energy of this set of signals is

$$E_{avg} = \frac{1}{8} (0 + 1 + 1 + 2 + 2 + 3 + 1 + 2) = \frac{12}{8} \text{ J}.$$

- Q3. Find a set of orthonormal basis functions for the following signal set. Find their vector forms and sketch their constellation.



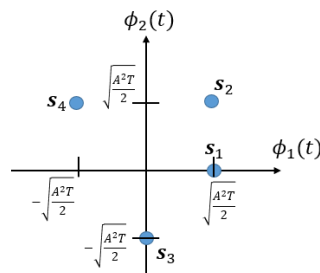
The answer is not unique. We start the G-S procedure from the signal $s_1(t)$.

1. $E_1 = \frac{1}{2}A^2T$. $\phi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} = \sqrt{\frac{2}{A^2T}} s_1(t)$ and $s_{11} = \sqrt{\frac{A^2T}{2}}$.
2. $s_{21} = \langle s_2(t), \phi_1(t) \rangle = \sqrt{\frac{A^2T}{2}}$. $g_2(t) = s_2(t) - s_{21}\phi_1(t) = s_2(t) - s_1(t)$ and $E_2 = \frac{1}{2}A^2T$. So $\phi_2(t) = \frac{g_2(t)}{\sqrt{E_2}} = \sqrt{\frac{2}{A^2T}} g_2(t)$ and $s_{22} = \sqrt{\frac{A^2T}{2}}$.
3. $s_{31} = \langle s_3(t), \phi_1(t) \rangle = 0$ and $s_{32} = \langle s_3(t), \phi_2(t) \rangle = -\sqrt{\frac{A^2T}{2}}$.
 $g_3(t) = s_3(t) - s_{31}\phi_1(t) - s_{32}\phi_2(t) = 0$ and no basis function from $s_3(t)$.
4. $s_{41} = \langle s_4(t), \phi_1(t) \rangle = -\sqrt{\frac{A^2T}{2}}$ and $s_{42} = \langle s_4(t), \phi_2(t) \rangle = \sqrt{\frac{A^2T}{2}}$.
 $g_4(t) = s_4(t) - s_{41}\phi_1(t) - s_{42}\phi_2(t) = 0$ and no basis function from $s_4(t)$.

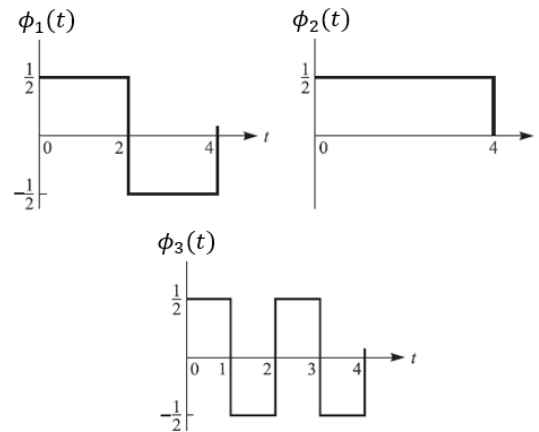
The set of basis functions are $\{\phi_1(t) = \sqrt{\frac{2}{A^2T}} s_1(t), \phi_2(t) = \sqrt{\frac{2}{A^2T}} g_2(t)\}$. The signal space is two dimensional and the vector forms of the signals are

$$s_1 = \left[\sqrt{\frac{A^2T}{2}}, 0 \right], s_2 = \left[\sqrt{\frac{A^2T}{2}}, \sqrt{\frac{A^2T}{2}} \right], s_3 = \left[0, -\sqrt{\frac{A^2T}{2}} \right] \text{ and } s_4 = \left[-\sqrt{\frac{A^2T}{2}}, \sqrt{\frac{A^2T}{2}} \right].$$

The constellation is shown below.



Q4. The waveforms for three functions $\phi_i(t)$, $1 \leq i \leq 3$, are given as below.



a) Show they are orthonormal.

By inspections, we have $\langle \phi_i(t), \phi_j(t) \rangle = \int_0^4 \phi_i(t) \phi_j^*(t) dt = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$ for $1 \leq i, j \leq 3$.

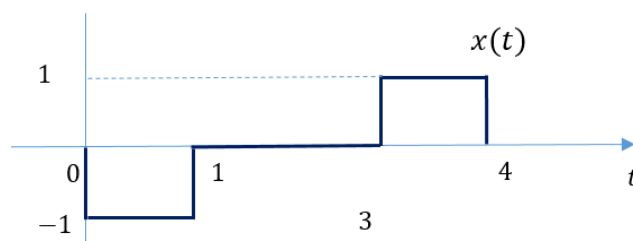
They are orthonormal.

b) Let the signal

$$x(t) = \begin{cases} -1, & 0 \leq t < 1 \\ 0, & 1 \leq t < 3 \\ 1, & 3 \leq t < 4 \end{cases}$$

1) Find the energy $\mathcal{E}_x$ of signal $x(t)$.

The plot of $x(t)$ is



$$\mathcal{E}_x = \langle x(t), x(t) \rangle = \int_0^4 x^2(t) dt = (-1)^2 \times 1 + 0^2 \times 2 + 1^2 \times 1 = 2.$$

2) Project $x(t)$ to the signal space constructed by $\phi_i(t)$, $1 \leq i \leq 3$. Determine the vector form $\mathbf{x}$ of $x(t)$.

Use $\{\phi_1(t), \phi_2(t), \phi_3(t)\}$ as the set of orthonormal basis functions. And $x(t)$ can be estimated as $\sum_{i=1}^3 x_i \phi_i(t)$ where $x_i = \int_0^4 x(t) \phi_i^*(t) dt = \int_0^4 x(t) \phi_i(t) dt$.

$$x_1 = \int_0^4 x(t) \phi_1(t) dt = -1 \times \frac{1}{2} + 0 \times \frac{1}{2} + 0 \times \left(-\frac{1}{2}\right) + 1 \times \left(-\frac{1}{2}\right) = -1.$$

$$x_2 = \int_0^4 x(t) \phi_2(t) dt = -1 \times \frac{1}{2} + 0 \times \frac{1}{2} + 0 \times \frac{1}{2} + 1 \times \frac{1}{2} = 0.$$

$$x_3 = \int_0^4 x(t) \phi_3(t) dt = -1 \times \frac{1}{2} + 0 \times \left(-\frac{1}{2}\right) + 0 \times \frac{1}{2} + 1 \times \left(-\frac{1}{2}\right) = -1.$$

$$\mathbf{x} = (-1, 0, -1) \text{ and } x(t) = \sum_{i=1}^3 x_i \phi_i(t).$$

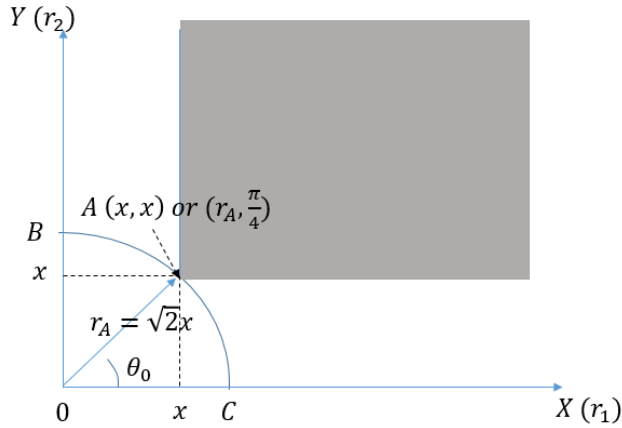
3) Find the energy of $x(t)$ using $\mathbf{x}$.

$$\begin{aligned} \langle x(t), x(t) \rangle &= \int_0^4 x^2(t) dt = \int_0^4 \sum_{i=1}^3 x_i \phi_i(t) \times \sum_{j=1}^3 x_j \phi_j(t) dt \\ &= \sum_{i=1}^3 x_i \left[\sum_{j=1}^3 x_j \left(\int_0^4 \phi_i(t) \phi_j(t) dt \right) \right] \\ &= \sum_{i=1}^3 x_i \times x_i = \langle \mathbf{x}, \mathbf{x} \rangle = 2. \end{aligned}$$

Q5. The Q function is defined as $Q(x) = \int_x^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$.

a) Show that $Q(x) \leq \frac{1}{2} e^{-\frac{x^2}{2}}$ when $x \geq 0$.

$$Q^2(x) = \left(\int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{r_1^2}{2}} dr_1 \right) \times \left(\int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{r_2^2}{2}} dr_2 \right) = \frac{1}{2\pi} \int_x^\infty \int_x^\infty e^{-\frac{r_1^2+r_2^2}{2}} dr_1 dr_2 \quad (*)$$



In a two-dimensional space, assume r_1 is along horizontal axis X and r_2 is along the vertical axis Y as shown in the figure above. The integration region in Eqn (*) is the shaded area. Note that this shaded area is smaller than the area above the arc BAC which is denoted by D. Also, since the integrand $e^{-\frac{r_1^2+r_2^2}{2}} \geq 0$ in both areas, we have

$$Q^2(x) = \frac{1}{2\pi} \int_x^\infty \int_x^\infty e^{-\frac{r_1^2+r_2^2}{2}} dr_1 dr_2 \leq \frac{1}{2\pi} \int_D e^{-\frac{r_1^2+r_2^2}{2}} dr_1 dr_2.$$

We then calculate the double integration on the right-hand side with polar coordinates.

Apply $dr_1 dr_2 = r d\theta dr$ and $r_1^2 + r_2^2 = r^2$. Since $x \geq 0$, $0 \leq \theta \leq \frac{\pi}{2}$ in the area D,

$$\begin{aligned} \frac{1}{2\pi} \int_D e^{-\frac{r_1^2+r_2^2}{2}} dr_1 dr_2 &= \frac{1}{2\pi} \int_0^{\frac{\pi}{2}} \int_{\sqrt{2}x}^\infty e^{-\frac{r^2}{2}} r dr d\theta = \frac{1}{2\pi} \int_0^{\frac{\pi}{2}} \int_{\sqrt{2}x}^\infty e^{-\frac{r^2}{2}} d(r^2/2) d\theta \\ &= \frac{1}{2\pi} \int_{\sqrt{2}x}^\infty e^{-\frac{r^2}{2}} d\theta \int_0^{\frac{\pi}{2}} d\theta d(r^2/2) = \frac{1}{4} \int_{\sqrt{2}x}^\infty e^{-\frac{r^2}{2}} d(r^2/2) = \frac{1}{4} e^{-x^2} \end{aligned}$$

From equation above, we have $Q^2(x) \leq \frac{1}{4} e^{-x^2}$ and $Q(x) \leq \sqrt{\frac{1}{4} e^{-x^2}} = \frac{1}{2} e^{-\frac{x^2}{2}}$.

b) For all $x \geq 0$, show that $\frac{x}{\sqrt{2\pi}(1+x^2)} e^{-\frac{x^2}{2}} < Q(x) < \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

Consider the integration $I(x) = \int_x^\infty e^{-\frac{t^2}{2}} t^{-2} dt$ for $x \geq 0$.

Since $e^{-\frac{t^2}{2}} t^{-2} \geq 0$ for $0 \leq x \leq t \leq \infty$, we have $I(x) > 0$ and

$$I(x) < \int_x^\infty e^{-\frac{t^2}{2}} x^{-2} dt = x^{-2} \int_x^\infty e^{-\frac{t^2}{2}} dt.$$

Let $u = e^{-\frac{t^2}{2}}$ and $v = -t^{-1}$,

$$uv|_x^\infty = -e^{-\frac{t^2}{2}} t^{-1}|_x^\infty = \frac{e^{-\frac{x^2}{2}}}{x}$$

$$\int_x^\infty u dv = \int_x^\infty e^{-\frac{t^2}{2}} t^{-2} dt = I(x)$$

$$\int_x^\infty v du = \int_x^\infty e^{-\frac{t^2}{2}} dt = \sqrt{2\pi} Q(x)$$

By integration by parts $uv = \int u dv + \int v du$,

$$\frac{e^{-\frac{x^2}{2}}}{x} = I(x) + \sqrt{2\pi} Q(x).$$

Since $I(x) > 0$, $\frac{e^{-\frac{x^2}{2}}}{x} > \sqrt{2\pi} Q(x)$ and $Q(x) < \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

Also $I(x) < x^{-2} \int_x^\infty e^{-\frac{t^2}{2}} dt = x^{-2} \sqrt{2\pi} Q(x)$,

$$\frac{e^{-\frac{x^2}{2}}}{x} < x^{-2} \sqrt{2\pi} Q(x) + \sqrt{2\pi} Q(x) = \sqrt{2\pi} (x^{-2} + 1) Q(x).$$

This shows $\frac{x}{\sqrt{2\pi}(1+x^2)} e^{-\frac{x^2}{2}} < Q(x)$.

Therefore, $\frac{x}{\sqrt{2\pi}(1+x^2)} e^{-\frac{x^2}{2}} < Q(x) < \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

Note that when x is a large value, $\frac{x}{\sqrt{2\pi}(1+x^2)} e^{-\frac{x^2}{2}} \approx \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ and $Q(x) \approx \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

Q6. Let random variable $X \sim \mathcal{N}(0, \sigma^2)$. If another random variable is generated by $Y = g(X)$. Find the PDF or PMF of Y for following cases.

$$\text{a) } g(x) = \begin{cases} -b, & x \leq -b \\ b, & x \geq b \\ x, & |x| < b \end{cases}$$

For this case, the new random variable Y is a mixed type random variable because $P[Y = -b] = P[X \leq -b]$ and $P[Y = b] = P[X \geq b]$ are nonzero probabilities.

The CDF for Y is $F_Y(y) = P[Y \leq y]$.

When $y < -b$, $F_Y(y) = P[Y \leq y] = 0$.

When $y = -b$, $F_Y(y) = P[Y \leq y] = P[Y = -b] = P[X \leq -b] = 1 - Q\left(-\frac{b}{\sigma}\right) = Q\left(\frac{b}{\sigma}\right)$.

When $|y| < b$, $F_Y(y) = P[Y \leq y] = P[X \leq y] = 1 - Q\left(\frac{y}{\sigma}\right)$.

When $y = b$, $F_Y(y) = P[Y \leq y] = P[Y \leq b] = 1$.

$$\text{CDF of } Y: \quad F_Y(y) = \begin{cases} 0, & y < -b; \\ 1 - Q\left(\frac{y}{\sigma}\right), & -b \leq y < b; \\ 1 & y \geq b. \end{cases}$$

$$\text{PDF of } Y: \quad f_Y(y) = \begin{cases} Q\left(\frac{b}{\sigma}\right) \delta(y + b), & y = -b \\ \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, & -b < y < b \\ Q\left(\frac{b}{\sigma}\right) \delta(y - b), & y = b \end{cases}$$

$$\text{b) } g(x) = \begin{cases} a, & x > 0 \\ 0, & x = 0 \\ b, & x < 0 \end{cases}$$

In this case, Y is a discrete random variable with support set $\{a, 0, b\}$. Its PMF is given as

$$P[Y = a] = P[X > 0] = Q(0) = 0.5$$

$$P[Y = 0] = P[X = 0] = 0$$

$$P[Y = b] = P[X < 0] = 1 - Q(0) = 0.5$$

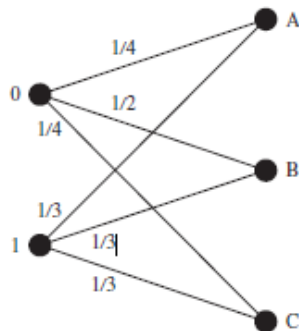
EE4402 Tutorial Two Solutions

Main topics in Chapter 3 and Chapter 4

1. Linear modulations.
 2. High dimensional modulations.
 3. Optimal receiver in AWGN channels.
 4. Optimal receiver performance in AWGN channels.
 5. Implementation of optimal receiver for AWGN channels.
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Q1. A communication channel with binary input and ternary output alphabets is shown in

Figure below. The probability of the input being 0 is 0.4. The transition probabilities due to the channel distortion are shown on the figure.



- a) For each of the channel outputs A, B and C , what is the best decision on channel input that minimizes the error probability?

For channel output A , the best decision rule should minimize the probability of detection error or maximize the probability of correct detection. The MAP rule is

$$\hat{s} = \arg_{s \in \{0,1\}} \max P[s|A].$$

$$P[A] = P[s = 0, A] + P[s = 1, A] = P[A|s = 0]P[s = 0] + P[A|s = 1]P[s = 1] = 0.3.$$

$$P[s = 0|A] = \frac{P[s=0,A]}{P[A]} = \frac{P[A|s=0]P[s=0]}{P[A]} = \frac{1}{3} \text{ and } P[s = 1|A] = 1 - P[s = 0|A] = \frac{2}{3}.$$

The detection rule $\hat{s} = \arg_{s \in \{0,1\}} \max P[s|A]$ should choose symbol 1 when A is received.

In a similar way, the MAP detection rule for B is choosing symbol 0 and 1 randomly because $P[s = 0|B] = P[s = 1|B] = 0.5$. The MAP detection rule for C is choosing symbol 1 when C is received because $P[s = 0|C] = 1/3$ and $P[s = 1|C] = 2/3$.

- b) If a 0 is transmitted and the optimal decision scheme derived in a) is applied at the receiver, what is the probability of error?

Denote the event symbol 0 is transmitted by $s = 0$, the detection error by e , the received symbol as r and detection output as d . If symbol 0 is transmitted, the event of detection error is the detection output is symbol 1.

$$\begin{aligned}
 P_{e|0} &= P[d = 1|s = 0] \\
 &= P[d = 1, r = A|s = 0] + P[d = 1, r = B|s = 0] + P[d = 1, r = C|s = 0] \\
 &= P[d = 1|r = A, s = 0]P[r = A|s = 0] + P[d = 1|r = B, s = 0]P[r = B|s = 0] \\
 &\quad + P[d = 1|r = C, s = 0]P[r = C|s = 0] \\
 &= P[d = 1|r = A]P[r = A|s = 0] + P[d = 1|r = B]P[r = B|s = 0] \\
 &\quad + P[d = 1|r = C]P[r = C|s = 0] \\
 &= 1 \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{2} + 1 \times \frac{1}{4} = \frac{3}{4}
 \end{aligned}$$

- c) What is the overall error probability for this channel if the optimal decision scheme is applied at the receiver?

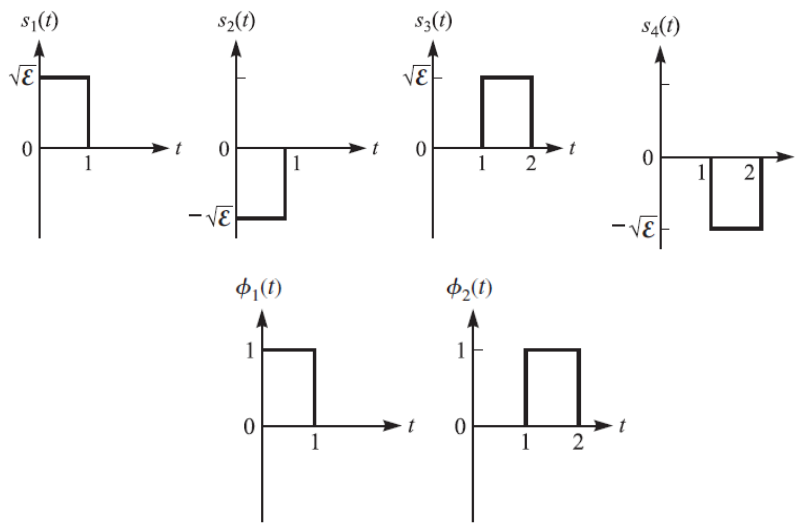
Apply the same idea in b),

$$P_{e|1} = 0 \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3} + 0 \times \frac{1}{3} = \frac{1}{6}$$

Therefore,

$$P_e = P_{e|0}P[s = 0] + P_{e|1}P[s = 1] = \frac{3}{4} \times 0.4 + \frac{1}{6} \times (1 - 0.4) = 0.4.$$

Q2. A modulation scheme uses the four signals $s_k(t)$, $k = 1, 2, 3, 4$, shown in the figure below, with the orthonormal functions $\phi_1(t)$ and $\phi_2(t)$ as the basis functions.



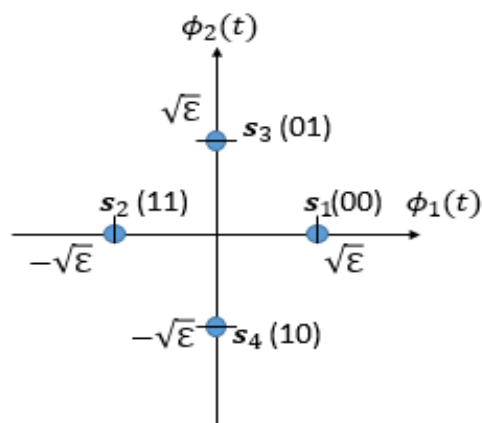
- a) Determine the vector forms of $s_k(t)$ for $k = 1, 2, 3, 4$.

The basis function of the signal set $\{s_i(t)\}$ can be found by GSO procedure. By inspection, we have $s_1(t) = \sqrt{\mathcal{E}}\phi_1(t)$, $s_2(t) = -\sqrt{\mathcal{E}}\phi_1(t)$, $s_3(t) = \sqrt{\mathcal{E}}\phi_2(t)$ and $s_4(t) = -\sqrt{\mathcal{E}}\phi_2(t)$.

Since there are two basis functions, the dimension $N = 2$. And $\mathbf{s}_1 = (\sqrt{\mathcal{E}}, 0)$,

$\mathbf{s}_2 = (-\sqrt{\mathcal{E}}, 0)$, $\mathbf{s}_3 = (0, \sqrt{\mathcal{E}})$ and $\mathbf{s}_4 = (0, -\sqrt{\mathcal{E}})$.

- b) Plot the signal constellation and label the signal points using Gray coding.



- c) What is this modulation based on the signal set?

Based on the constellation, it is a QPSK modulation.

- d) Find the minimum Euclidean distance, average signal energy and average bit energy?

The minimum distance between symbols in the constellation is $\sqrt{2\mathcal{E}}$. Since all the symbols have the same amplitude, they have the same symbol energy $\mathcal{E}$. Since $M = 4$, each symbol can carry $\log_2 M = 2$ bit information and the average bit energy is $\mathcal{E}/2$.

- e) If the signal set is transmitted in an AWGN channel and all signals are equiprobable, what is the optimal detection rule?

Since all symbols are equiprobable, the optimal detection rule is the maximum likelihood detection which is

$$\hat{s} = \arg \max_{1 \leq i \leq 4} P[\mathbf{r}|\mathbf{s}_i]$$

In an AWGN channel with each noise sample variance $N_0/2$, $\mathbf{r} = \mathbf{s} + \mathbf{n}$. For a given $\mathbf{s} = \mathbf{s}_i$,

$\mathbf{n} = \mathbf{r} - \mathbf{s}_i$ and $P[\mathbf{r}|\mathbf{s}_i] = P[\mathbf{n} = \mathbf{r} - \mathbf{s}_i] = \frac{1}{\pi N_0} e^{-\frac{\|\mathbf{r} - \mathbf{s}_i\|^2}{N_0}}$. So the ML rule is

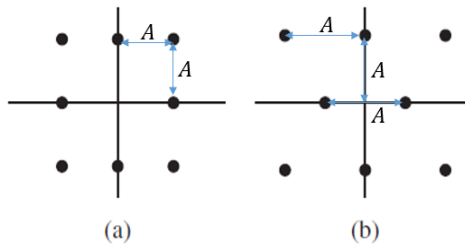
$$\hat{s} = \arg \max_{1 \leq i \leq 4} P[\mathbf{r}|\mathbf{s}_i] = \arg \max_{1 \leq i \leq 4} \frac{1}{\pi N_0} e^{-\frac{\|\mathbf{r} - \mathbf{s}_i\|^2}{N_0}} = \arg \max_{1 \leq i \leq 4} e^{-\frac{\|\mathbf{r} - \mathbf{s}_i\|^2}{N_0}} = \arg \min_{1 \leq i \leq 4} \|\mathbf{r} - \mathbf{s}_i\|^2.$$

The decision rule can be further simplified as

$$\hat{s} = \arg \min_{1 \leq i \leq 4} \left(\|\mathbf{r}\|^2 + \langle \mathbf{s}_i, \mathbf{s}_i \rangle - 2\langle \mathbf{r}, \mathbf{s}_i \rangle \right) = \arg \max_{1 \leq i \leq 4} \langle \mathbf{r}, \mathbf{s}_i \rangle,$$

because $\langle \mathbf{s}_i, \mathbf{s}_i \rangle = \mathcal{E}$ for $1 \leq i \leq 4$ and the term $\|\mathbf{r}\|^2$ can be dropped.

Q3. Consider the two 8-point QAM signal constellations shown in figure below. The minimum distance between adjacent points is A which is a positive constant.



- a) Find the vector form of each signal in two constellations.

Since all the symbols are in the two-dimensional space, the symbol space dimension $N = 2$.

By simple geometry, the vector form for symbols in the constellation of (a) are

$$(A, A), (A, 0), (A, -A), (-A, 0), (-A, -A), (-A, 0), (-A, A), (0, A)$$

and the vector form for symbols in constellation of (b) are

$$(A, A), (A/2, 0), (A, -A), (-A, 0), (-A, -A), (-A/2, 0), (-A, A), (0, A).$$

- b) If all signals are equiprobable, determine the average signal energy for each constellation and indicate which constellation is more power-efficient?

There are 8 symbols in each constellation and all the symbols are transmitted with equal probability $\frac{1}{8}$. The average energy of constellation is

$$E_{avg} = \frac{1}{8} \sum_{1 \leq i \leq 8} ||s_i||^2.$$

$$\text{In constellation (a), } E_{avg} = \frac{1}{8} (2A^2 \times 4 + A^2 \times 4) = \frac{3}{2} A^2.$$

$$\text{In constellation (b), } E_{avg} = \frac{1}{8} \left(2A^2 \times 4 + A^2 \times 2 + \frac{A^2}{4} \times 2 \right) = 1.3125 A^2$$

Both constellations have the same minimum Euclidean distance. But constellation (b) needs less average energy to achieve it and it is more power efficient than the constellation (a). If both constellations have the same average energy, the constellation (b) will have a larger minimum Euclidean distance than that of constellation (a).

- c) If the communication is in an AWGN channel with noise variance $\frac{N_0}{2}$, estimate the symbol error rate using the union bound and simplify the union bound in terms of minimum Euclidean distance.

Denote the pairwise symbol error probability between symbol s_m and symbol s_k by $P_{m \rightarrow k}$ and the Euclidean distance between symbol s_m and symbol s_k by d_{mk} . The union bound of symbol error rate is

$$\frac{1}{M} \sum_{1 \leq m \leq M} \sum_{\substack{1 \leq k \leq M \\ k \neq m}} P_{m \rightarrow k} = \frac{1}{M} \sum_{1 \leq m \leq M} \sum_{\substack{1 \leq k \leq M \\ k \neq m}} Q \left(\sqrt{\frac{d_{mk}^2}{2N_0}} \right)$$

It can be simplified by replacing the Euclidean distance d_{mk} by the minimum Euclidean distance of the constellation d_{min} . The union bound becomes

$$\frac{1}{M} \sum_{1 \leq m \leq M} \sum_{\substack{1 \leq k \leq M \\ k \neq m}} Q \left(\sqrt{\frac{d_{mk}^2}{2N_0}} \right) \leq \frac{1}{M} \sum_{1 \leq m \leq M} \sum_{\substack{1 \leq k \leq M \\ k \neq m}} Q \left(\sqrt{\frac{d_{min}^2}{2N_0}} \right)$$

The right-hand side can be simplified for the given constellations as

$$\frac{1}{M} \sum_{1 \leq m \leq M} (M-1) Q \left(\sqrt{\frac{d_{min}^2}{2N_0}} \right) = (M-1) Q \left(\sqrt{\frac{d_{min}^2}{2N_0}} \right) = (M-1) Q \left(\sqrt{\frac{A^2}{2N_0}} \right)$$

The symbol signal-to-noise ratio in this problem can be defined as $\gamma_s = \frac{E_s}{N_0}$.

For constellation (a), $E_s = 1.5A^2$ and $\frac{A^2}{2N_0} = \frac{E_s}{3N_0} = \frac{\gamma_s}{3}$. The corresponding union bound is

$$(M-1) Q \left(\sqrt{\frac{\gamma_s}{3}} \right).$$

For constellation (b), $E_s = 1.3125A^2$ and $\frac{A^2}{2N_0} = \frac{E_s}{2.625N_0} = \frac{\gamma_s}{2.625}$. The corresponding union

$$\text{bound is } (M-1) Q \left(\sqrt{\frac{\gamma_s}{2.625}} \right).$$

Q4. A binary digital communication system employs the signals

$$s_0(t) = 0, \quad 0 \leq t \leq T_s$$

$$s_1(t) = A, \quad 0 \leq t \leq T_s$$

for transmitting the information. This is called on-off keying (OOK) signaling. The demodulator cross correlates the received signal $r(t)$ with $s_0(t)$ and $s_1(t)$, respectively, and samples the output of the correlator at $t = T_s$.

a) If all signals are equiprobable, determine the optimum detector for an AWGN channel.

The OOK signal is with dimension $N = 1$. The energy of signal $s_0(t)$ is 0 and the energy of $s_1(t)$ is $\mathcal{E}_1 = \int_0^{T_s} |s_1(t)|^2 dt = A^2 T_s$. Choose the basis function $\phi(t) = \frac{s_1(t)}{\sqrt{\mathcal{E}_1}} = \frac{s_1(t)}{A\sqrt{T_s}}$. The vector form of $s_0(t)$ and $s_1(t)$ are $s_0 = 0$ and $s_1 = A\sqrt{T_s}$.

Assume the noise sample variance is $\frac{N_0}{2}$. Since the received signal $r = s + n$,

$$P[r|s_0] = P[n = r - s_0] = P[n = r] = \frac{1}{\sqrt{\pi N_0}} e^{-\frac{r^2}{N_0}}$$

$$P[r|s_1] = P[n = r - s_1] = P[n = r - A\sqrt{T_s}] = \frac{1}{\sqrt{\pi N_0}} e^{-\frac{(r - A\sqrt{T_s})^2}{N_0}}$$

The decision rule can be

$$\frac{P[r|s_0]}{P[r|s_1]} = e^{\frac{(r - A\sqrt{T_s})^2 - r^2}{N_0}} \underset{s_1}{\overset{s_0}{\geq}} 1$$

or

$$(r - A\sqrt{T_s})^2 - r^2 = -(2r - A\sqrt{T_s})A\sqrt{T_s} \underset{s_1}{\overset{s_0}{\geq}} 0.$$

Because $A\sqrt{T_s} > 0$, the decision rule can be further simplified as

$$r \underset{s_1}{\overset{s_0}{\leq}} \frac{A\sqrt{T_s}}{2}.$$

This result can also be derived based on the Euclidean distance of s_0 and s_1 which is $A\sqrt{T_s}$. Its middle point is $\frac{A\sqrt{T_s}}{2}$. If $r \geq \frac{A\sqrt{T_s}}{2}$, it is detected as s_1 and otherwise it will be detected as s_0 .

- b) Determine the probability of symbol error as a function of the $\frac{E_b}{N_0}$ if both signals are transmitted with equal probability.

$$P_e = P[e, s_0] + P[e, s_1] = P[e|s_0]P[s_0] + P[e|s_1]P[s_1] = \frac{1}{2}P[e|s_0] + \frac{1}{2}P[e|s_1].$$

$$P[e|s_0] = \int_{\frac{A\sqrt{T_s}}{2}}^{\infty} \frac{1}{\sqrt{\pi N_0}} e^{-\frac{r^2}{N_0}} dr = \int_{\frac{A\sqrt{T_s}}{2}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{r^2}{2}} dr = Q\left(\sqrt{\frac{\frac{A^2 T_s}{2}}{N_0}}\right).$$

By symmetry, $P[e|s_1] = P[e|s_0] = Q\left(\sqrt{\frac{\frac{A^2 T_s}{2}}{N_0}}\right)$ and $P_e = Q\left(\sqrt{\frac{\frac{A^2 T_s}{2}}{N_0}}\right)$. Note that $\frac{A^2 T_s}{2}$ is the

average energy of the symbols and $\frac{A^2 T_s}{2 N_0}$ is the signal-to-noise ratio (SNR)

This result can also be derived from the pair-wise SER directly as $P_e = Q\left(\sqrt{\frac{d_{0,1}^2}{2N_0}}\right) = Q\left(\sqrt{\frac{A^2 T_s}{2N_0}}\right)$.

- c) Compare performance of the on-off signaling and the antipodal signaling. (Assume both are with equiprobable signals and compare their SER vs. $\frac{E_s}{N_0}$.)

The antipodal signaling SER is $P_e = Q\left(\sqrt{\frac{2\mathcal{E}_A}{N_0}}\right)$, where the average signal energy is $\mathcal{E}_A$.

For the OOK signaling, the average signal energy is $\mathcal{E}_O = \frac{A^2 T_s}{2}$ and $P_e = Q\left(\sqrt{\frac{\mathcal{E}_O}{N_0}}\right)$.

To achieve the same SER, $\mathcal{E}_O = 2\mathcal{E}_A$. That is OOK needs 3dB more energy than antipodal signaling and antipodal is more power efficient than OOK signaling.

Q5. A ternary communication system transmits one of three real equiprobable signals $s(t)$, 0, or $-s(t)$ every T seconds. The received signal is $r(t) = s(t) + z(t)$, $r(t) = z(t)$, or $r(t) = -s(t) + z(t)$, where $z(t)$ is white Gaussian noise with sample variance $N_0/2$. The energy of $s(t)$ is E_s . The optimum receiver computes the correlation metric

$$U = \int_0^T r(t)s(t)dt$$

and compares U with a threshold $A = \frac{E_s}{2}$ and a threshold $-A$. If $U > A$, the decision is made that $s(t)$ was sent. If $U < -A$, the decision is made in favor of $-s(t)$. If $-A < U < A$, the decision is made in favor of 0. Note that $N = \int_0^T z(t)s(t)dt$ is a zero mean Gaussian random variable of variance $\frac{E_s N_0}{2}$.

- a) Determine the three conditional probabilities of error: P_e given that $s(t)$ was sent, P_e given that $-s(t)$ was sent, and P_e given that 0 was sent.

If the symbol $s(t)$ is sent, the detection metric

$$U = \int_0^T r(t)s(t)dt = \int_0^T (s(t) + z(t))s(t)dt = \int_0^T s(t)s(t)dt + \int_0^T z(t)s(t)dt = E_s + N.$$

$$P[e|s(t)] = P[U \leq A] = P[E_s + N \leq A] = P[N \geq E_s - A] = \int_{E_s - A}^{\infty} \frac{1}{\sqrt{\pi E_s N_0}} e^{-\frac{n^2}{E_s N_0}} dn$$

$$= \int_{\frac{E_s - A}{\sqrt{\frac{E_s N_0}{2}}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{n}{\sqrt{\frac{E_s N_0}{2}}}\right)^2}{2}} d\left(\frac{n}{\sqrt{\frac{E_s N_0}{2}}}\right) = Q\left(\frac{E_s - A}{\sqrt{\frac{E_s N_0}{2}}}\right)$$

If the symbol $-s(t)$ is sent, the detection metric $U = -E_s + N$.

$$P[e|-s(t)] = P[U \geq -A] = P[N \geq E_s - A] = Q\left(\frac{E_s - A}{\sqrt{\frac{E_s N_0}{2}}}\right)$$

If the symbol 0 is sent, the detection metric $U = N$

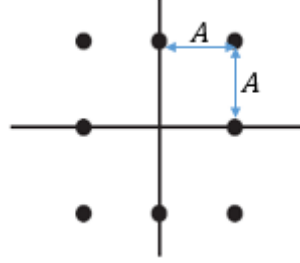
$$P[e|0] = P[\{U > A\} \cup \{U < -A\}] = P[N > A] + P[N < -A] = 2Q\left(\frac{A}{\sqrt{\frac{E_s N_0}{2}}}\right).$$

- b) Determine the average probability of error P_e as a function of the threshold A , assuming that these three symbols are equiprobable.

$$P_e = P[e|s(t)]P[s(t)] + P[e|-s(t)]P[-s(t)] + P[e|0]P[0]$$

$$= \frac{2}{3} \left(Q\left(\frac{E_s - A}{\sqrt{\frac{E_s N_0}{2}}}\right) + Q\left(\frac{A}{\sqrt{\frac{E_s N_0}{2}}}\right) \right).$$

Q6. The signal constellation for a communication system with 8 equiprobable symbols is shown in figure below. The channel is AWGN with noise variance $N_0/2$.



- a) Using the union bound, find a bound in terms of A and N_0 on the error probability for this channel.

The minimum Euclidean distance $d_{min} = A$ and the union bound for the SER is

$$(M - 1)Q\left(\sqrt{\frac{d_{min}^2}{2N_0}}\right) = 7Q\left(\sqrt{\frac{A^2}{2N_0}}\right).$$

- b) Determine the average SNR per bit for this channel.

The vector form of all symbols are

$$(A, 0), (A, A), (0, A), (-A, A), (-A, 0), (-A, -A), (0, -A), (A, -A).$$

The average energy for the given constellation is

$$\mathcal{E}_{avg} = \frac{1}{8} \sum_{i=1}^8 \mathcal{E}_i = \frac{1}{8} \sum_{i=1}^8 \langle s_i, s_i \rangle = \frac{1}{8} \times 12A^2 = \frac{3}{2}A^2.$$

Since each symbol represents $k = \log_2 M = 3$ bits. The average bits energy is

$$\mathcal{E}_{bavg} = \frac{\mathcal{E}_{avg}}{k} = \frac{1}{2}A^2.$$

The average SNR per bit is $\gamma_b = \frac{\mathcal{E}_{bavg}}{N_0} = \frac{A^2}{2N_0}$.

- c) Express the bound found in a) in terms of the average SNR per bit.

The union bound can be expressed as $7Q(\sqrt{\gamma_b})$ from results in a) and b).

Q7. Two signals $s_1(t) = 3d \cos(2\pi f_0 t)$, $s_2(t) = 4d \sin(2\pi f_0 t)$, where $0 \leq t \leq T_s$, are used as the signal set for digital transmission in an AWGN channel with noise variance $\frac{N_0}{2}$. T_s is the symbol duration. Assume $T_s \times f_0$ is an integer and two signals are transmitted with equal probability.

- (a) Indicate the basis function(s) of the signal set and their vector representations under this set of basis function(s).

The energy of the $s_1(t)$ and $s_2(t)$ are

$$\varepsilon_1 = \langle s_1(t), s_1(t) \rangle = \int_0^{T_s} 9d^2 \cos^2(2\pi f_0 t) dt = 4.5d^2 T_s,$$

and

$$\varepsilon_2 = \langle s_2(t), s_2(t) \rangle = \int_0^{T_s} 16d^2 \sin^2(2\pi f_0 t) dt = 8d^2 T_s.$$

It is also easy to see that $s_1(t)$ and $s_2(t)$ are orthogonal because

$$\langle s_1(t), s_2(t) \rangle = \int_0^{T_s} 12d^2 \sin(2\pi f_0 t) \cos(2\pi f_0 t) dt = \int_0^{T_s} 6d^2 \sin(4\pi f_0 t) dt = 0.$$

So the basis function set can be

$$\phi_1(t) = \frac{s_1(t)}{\sqrt{\varepsilon_1}} = \sqrt{\frac{2}{T_s}} \cos(2\pi f_0 t) \text{ and } \phi_2(t) = \frac{s_2(t)}{\sqrt{\varepsilon_2}} = \sqrt{\frac{2}{T_s}} \sin(2\pi f_0 t) \text{ for } 0 \leq t < T_s.$$

Project $s_1(t)$ and $s_2(t)$ to the basis set, we have $\mathbf{s}_1 = \left(3d \sqrt{\frac{T_s}{2}}, 0 \right)$ and $\mathbf{s}_2 = \left(0, 4d \sqrt{\frac{T_s}{2}} \right)$.

- (b) Find the average bit energy and the signal-to-noise ratio $\gamma_b = E_b/N_0$.

Since the constellation size $M = 2$, each symbol can be represented by $k = 1$ bit and $E_b =$

$\varepsilon_s = 0.5(\varepsilon_1 + \varepsilon_2) = 6.25d^2 T_s$. So we have

$$\gamma_b = \frac{E_b}{N_0} = \frac{6.25d^2 T_s}{N_0}.$$

- (c) Derive the decision boundary for the receiver to minimize the bit error probability and show it in a sketch of the constellation.

Since $s_1(t)$ and $s_2(t)$ are transmitted with equal probability in an AWGN channel, the optimal decision rule is the ML rule.

Denote the received signal in vector form as $\mathbf{r} = (r_1, r_2)$. Based on the ML rule,

$$\begin{aligned} D_1 &= \{\mathbf{r} | P[\mathbf{r} | \mathbf{s}_1] > P[\mathbf{r} | \mathbf{s}_2]\} = \{\mathbf{r} | \|\mathbf{r} - \mathbf{s}_1\| < \|\mathbf{r} - \mathbf{s}_2\|\} \\ &= \{\mathbf{r} | \langle \mathbf{s}_1, \mathbf{s}_1 \rangle - 2\langle \mathbf{r}, \mathbf{s}_1 \rangle < \langle \mathbf{s}_2, \mathbf{s}_2 \rangle - 2\langle \mathbf{r}, \mathbf{s}_2 \rangle\} \\ &= \{\mathbf{r} | \varepsilon_1 - 2\langle \mathbf{r}, \mathbf{s}_1 \rangle < \varepsilon_2 - 2\langle \mathbf{r}, \mathbf{s}_2 \rangle\} \end{aligned}$$

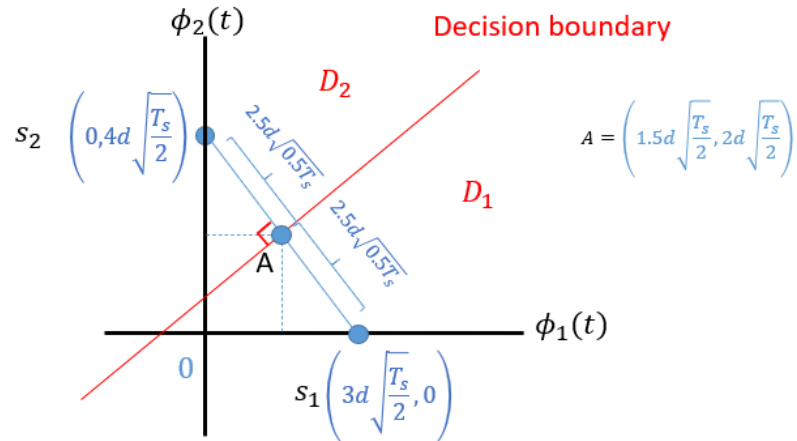
Since $\langle \mathbf{r}, \mathbf{s}_1 \rangle = 3d\sqrt{\frac{T_s}{2}}r_1$ and $\langle \mathbf{r}, \mathbf{s}_2 \rangle = 4d\sqrt{\frac{T_s}{2}}r_2$,

$$D_1 = \left\{ \mathbf{r} \left| 4.5d^2T_s - 6d\sqrt{\frac{T_s}{2}}r_1 < 8d^2T_s - 8d\sqrt{\frac{T_s}{2}}r_2 \right. \right\} = \left\{ \mathbf{r} \left| 4r_2 - 3r_1 < 1.75d\sqrt{2T_s} \right. \right\}$$

Since only two symbols in the constellation, $D_2 = D_1^C$. The constellation is 2-dimensional and

the decision boundary is shown as below where Point A has coordinates $\left(1.5d\sqrt{\frac{T_s}{2}}, 2d\sqrt{\frac{T_s}{2}}\right)$ and

the red line indicate the decision boundary.



(d) Calculate the bit error rate and express it using Q function and γ_b .

When $\mathbf{s}_1$ is transmitted, the detection error probability is the pair-wise symbol error

probability. The Euclidean distance between $\mathbf{s}_1$ and $\mathbf{s}_2$ is $d_{12} = \sqrt{4.5d^2T_s + 8d^2T_s} = 5d\sqrt{\frac{T_s}{2}}$.

So

$$\begin{aligned} P_e &= P[\mathbf{s}_1]P_{e|\mathbf{s}_1} + P[\mathbf{s}_2]P_{e|\mathbf{s}_2} = \frac{1}{2}[P_{e|\mathbf{s}_1} + P_{e|\mathbf{s}_2}] \\ &= P_{e|\mathbf{s}_1} = Q\left(\sqrt{\frac{d_{12}^2}{2N_0}}\right) = Q\left(\sqrt{\frac{25d^2T_s/2}{2N_0}}\right) = Q(\sqrt{\gamma_b}). \end{aligned}$$

Q8. The discrete sequence

$$r_k = \sqrt{\mathcal{E}}c_k + n_k, k = 1, 2, \dots, n$$

represents the output sequence of samples from a demodulator, where $c_k = \pm 1$ are elements

in one of two possible codewords, $\mathbf{c}_1 = [1 \ 1 \ \dots \ 1]$ and $\mathbf{c}_2 = [1 \ 1 \ \dots \ 1 \ -1 \ \dots \ -1]$. The code word $\mathbf{c}_2$ has w elements that are $+1$ and $n - w$ elements that are -1 , where w is a positive integer or zero. The noise sequence $\{n_k\}$ is white Gaussian with variance $\frac{N_0}{2}$. Assume two codewords are equiprobable.

a) What is the optimum maximum-likelihood detector for the two possible transmitted signals?

Define two symbols $\mathbf{s}_1 = \sqrt{\mathcal{E}}\mathbf{c}_1$ and $\mathbf{s}_2 = \sqrt{\mathcal{E}}\mathbf{c}_2$. The n -dimensional orthonormal basis set are selected as $\phi_i = (0, 0, \dots, 0, \underbrace{1}_{i^{th}}, 0, 0, \dots, 0)$ for $1 \leq i \leq n$.

In vector form, $\mathbf{s}_1 = (\underbrace{\sqrt{\mathcal{E}}, \sqrt{\mathcal{E}}, \dots, \sqrt{\mathcal{E}}}_n)$ and $\mathbf{s}_2 = (\underbrace{\sqrt{\mathcal{E}}, \sqrt{\mathcal{E}}, \dots, \sqrt{\mathcal{E}}}_w, \underbrace{-\sqrt{\mathcal{E}}, \dots, -\sqrt{\mathcal{E}}}_{n-w})$. Denote received signal vector form $\mathbf{r} = (r_1, r_2, \dots, r_n)$.

Since two code words are equiprobable, the optimal detection rule is the ML detection rule

$$\hat{s} = \arg \min_{s \in \{\mathbf{s}_1, \mathbf{s}_2\}} \|\mathbf{r} - \mathbf{s}\|^2.$$

If $\mathbf{s}_1$ is sent, the metric is $\sum_{i=1}^n (r_i - \sqrt{\mathcal{E}})^2$.

If $\mathbf{s}_2$ is sent, the metric is $\sum_{i=1}^w (r_i - \sqrt{\mathcal{E}})^2 + \sum_{i=w+1}^n (r_i + \sqrt{\mathcal{E}})^2$.

The metric can be simplified as $\sum_{i=w+1}^n (r_i - \sqrt{\mathcal{E}})^2$ and $\sum_{i=w+1}^n (r_i + \sqrt{\mathcal{E}})^2$ because the first w terms are the same. Since

$$\sum_{i=w+1}^n (r_i - \sqrt{\mathcal{E}})^2 = \sum_{i=w+1}^n (r_i^2 - 2\sqrt{\mathcal{E}}r_i + \mathcal{E}) = \sum_{i=w+1}^n r_i^2 - 2\sqrt{\mathcal{E}} \sum_{i=w+1}^n r_i + \mathcal{E}(n - w)$$

and

$$\sum_{i=w+1}^n (r_i + \sqrt{\mathcal{E}})^2 = \sum_{i=w+1}^n (r_i^2 + 2\sqrt{\mathcal{E}}r_i + \mathcal{E}) = \sum_{i=w+1}^n r_i^2 + 2\sqrt{\mathcal{E}} \sum_{i=w+1}^n r_i + \mathcal{E}(n-w).$$

The rule becomes

$$-\sum_{i=w+1}^n r_i \underset{\mathbf{s}_2}{\overset{\mathbf{s}_1}{\leq}} \sum_{i=w+1}^n r_i \text{ or } \sum_{i=w+1}^n r_i \underset{\mathbf{s}_2}{\overset{\mathbf{s}_1}{\geq}} 0.$$

b) Determine the probability of error as a function of the parameters $N_0, \mathcal{E}, w$ and n .

The Euclidean distance between these two symbols is $d_{12} = |\mathbf{s}_1 - \mathbf{s}_2| = 2\sqrt{\mathcal{E}(n-w)}$. Their SER is their pair-wise SER which is

$$P_e = Q\left(\sqrt{\frac{d_{12}^2}{2N_0}}\right) = Q\left(\sqrt{\frac{4\mathcal{E}(n-w)}{2N_0}}\right) = Q\left(\sqrt{\frac{2\mathcal{E}(n-w)}{N_0}}\right).$$

c) What is the value of w that minimizes the error?

Since $Q(x)$ is decreasing respect to x , to minimize the SER, w should be minimized to its minimum value 0.

EE4402 Tutorial Three Solutions

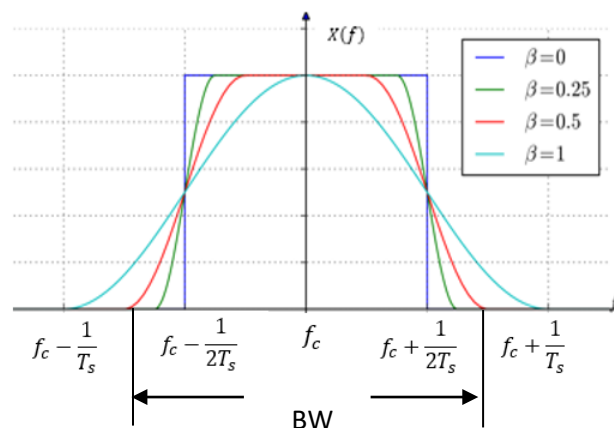
Main topics in Chapter 5- 7

1. Signaling bandlimited channel.
 2. Linear block codes.
 3. Convolutional codes.
-

1. An 8-kHz bandpass channel is to be used for transmission of data at a rate of 9600 bits/s. If the additive zero-mean Gaussian noise sample variance is $\frac{1}{2}N_0 = 10^{-10}$ in the channel, design following modulations and determine the average power that achieves a bit error probability of 10^{-6} using a signal pulse with a raised cosine spectrum having a roll-off factor of at least 50 percent.
 - a) PAM.
 - b) PSK.

Assume the modulation is M-ary and symbol duration is T_s . To transmit the signal at 9600 bits/s within 8 kHz (**bandpass**) without ISI using a raised cosine pulse with roll-off factor β , we should have

$$X(f) = \begin{cases} T_s, & |f| \leq \frac{1-\beta}{2T_s}, \\ \frac{T_s}{2} \left(1 + \cos \left(\frac{\pi T_s}{\beta} \left[|f| - \frac{1-\beta}{2T_s} \right] \right) \right), & \frac{1-\beta}{2T_s} < |f| \leq \frac{1+\beta}{2T_s}, \\ 0, & |f| > \frac{1+\beta}{2T_s}, \end{cases}$$

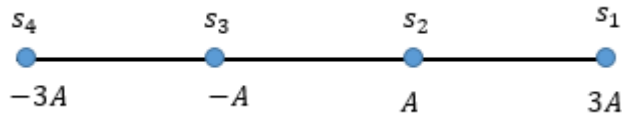


and

$$8000 > \frac{1+\beta}{T_s} \text{ and } \frac{1}{T_s} \times \log_2 M = 9600.$$

Combining above two, we have $\log_2 M > 1.2(1 + \beta)$. The smallest M satisfying the constraint $\beta > 0.5$ is $M = 4$ and we have $T_s = \frac{1}{4800} s$. So we need to consider 4-ary modulation for both cases.

a) For 4-PAM, assume 4 signals are $s_1 = 3A, s_2 = A, s_3 = -A, s_4 = -3A$ and equiprobable.



Its average energy is

$$E_{avg} = \frac{9A^2T_s + A^2T_s}{2} = 5A^2T_s = \frac{5}{4800}A^2.$$

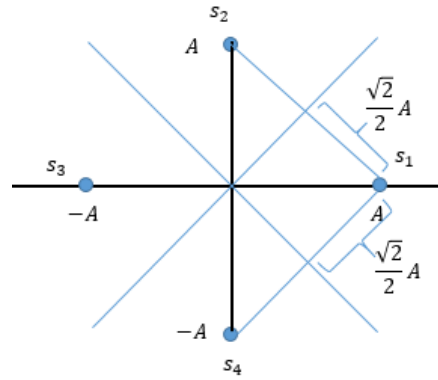
Disregarded the coding of the constellation, by symmetry, its BER is upper bounded by its symbol error rate P_e (overestimated by assuming each symbol error leads to 2 error bits.)

$$P_{eb} \approx P_e = 2(P_{e|s_1} \times P[s_1] + P_{e|s_2} \times P[s_2]) = 1.5Q\left(\sqrt{\frac{2A^2}{N_0}}\right)$$

where $P_{e|s_1} = Q\left(\sqrt{\frac{2A^2}{N_0}}\right)$, $P_{e|s_2} = 2Q\left(\sqrt{\frac{2A^2}{N_0}}\right)$ and $P[s_1] = P[s_2] = 1/4$.

Since $P_{eb} = 1.5Q\left(\sqrt{\frac{2A^2}{N_0}}\right) = 10^{-6}$, by the inverse Q-function, $\sqrt{\frac{2A^2}{N_0}} = 4.8347$ and $A = 4.8347 \times 10^{-5}$. Therefore, $E_s = \frac{5}{4800} \times A^2 = 2.4348 \times 10^{-12} \text{ joules}$ will ensure the bit error probability is at most 10^{-6} for 4-PAM.

b) Another case is QPSK. Assume Gray mapping is applied.



By symmetry, the BER is

$$\begin{aligned}
 P_{eb} &= \frac{1}{2} \times 4 \times [P[s_1 \rightarrow s_2] + P[s_1 \rightarrow s_4] + 2P[s_1 \rightarrow s_3]] \times P[s_1] \\
 &= Q\left(\sqrt{\frac{A^2}{N_0}}\right) \left(1 - Q\left(\sqrt{\frac{A^2}{N_0}}\right)\right) + \left(Q\left(\sqrt{\frac{A^2}{N_0}}\right)\right)^2 \\
 &= Q\left(\sqrt{\frac{A^2}{N_0}}\right)
 \end{aligned}$$

Note that this is the exact BER.

Solving for A from the equation $Q\left(\sqrt{\frac{A^2}{N_0}}\right) = 10^{-6}$, we have $A = 6.7224 \times 10^{-5}$. The minimum symbol energy $E_{avg} = A^2 T_s = \frac{A^2}{4800} \approx 9.4146 \times 10^{-13} \text{ joules}$ is needed to achieve the bit error probability 10^{-6} .

2. Let $\mathcal{C}$ be a binary $(6, 3)$ code with parity check matrix,

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

a) What is the minimum Hamming distance of this code?

The minimum Hamming distance is the same as the minimum Hamming weight of all nonzero code words. For $\mathbf{c} \in \mathcal{C}$ with the minimum Hamming weight, since

$\mathbf{c} \times H^T = \mathbf{0}$, it indicates the minimum number of linear dependent columns of H . By simple inspection of H , we have $d_{min} = 3$.

b) Find the generator matrix G .

The generator matrix is a 3×6 binary matrix with its rows orthogonal to the rows of H . Based on the relationship of G and H , we have

$$G = [P|I_{3 \times 3}] \text{ and } P = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}. \text{ That is } G = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

c) Encode message $\mathbf{m} = (1 \ 0 \ 1)$.

$$\mathbf{c} = \mathbf{m} \times G = (1 \ 0 \ 1) \times \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} = (1 \ 0 \ 1 \ 1 \ 0 \ 1)$$

d) Suppose the code is used for error detection over a binary symmetric channel with crossover probability $p = 10^{-3}$. Find the probability of undetected error.

Since the generator matrix G has full rank 3. There are 2^3 code words in this code which are listed below.

Message	Code word	Hamming weight	Message	Code word	Hamming weight
000	000000	0	110	110110	4
100	011100	3	101	101101	4
010	101010	3	011	011011	4
001	110001	3	111	000111	3

Besides the all-zero code word, there are 4 code words with Hamming weight 3 and 3 code words with Hamming weight 4.

If an undetected error occurs, the error pattern must be a valid code word other than the all-zero code word. Therefore, the probability of these error patterns in the BSC channel is

$$\begin{aligned}
 P_{undetected} &= 4p^3 (1 - p)^3 + 3p^4(1 - p)^2 \\
 &= 4 \times 10^{-9}(0.999)^3 + 3 \times 10^{-12}(0.999)^2 \approx 4 \times 10^{-9}.
 \end{aligned}$$

e) Decode the code word $\mathbf{c} = (\times \times \times 1 1 0)$ where $\times$ means an unknown value.

Denote the first three coordinates as c_1, c_2 and c_3 in the code word. Since $\mathbf{c} \times \mathbf{H}^T = \mathbf{0}$ for $\mathbf{c} \in \mathcal{C}$, we have $(c_1 \ c_2 \ c_3 \ 1 \ 1 \ 0) \times \mathbf{H}^T = \mathbf{0}$. That is

$$c_1 + 1 = 0$$

$$c_2 + 1 = 0$$

$$c_3 + 1 + 1 = 0$$

Note the addition are binary XOR operation. Therefore, $c_1 = 1, c_2 = 1$ and $c_3 = 0$.

The received noisy vector $\mathbf{r} = (\times \times \times 1 1 0)$ is decoded as the code word $(1 1 0 1 1 0)$.

3. Let $\mathcal{C}$ be the binary repetition code of length 5.

a) List the vectors which will be decoded as 11111 using Hamming distance rule.

The Hamming distance rule decode a received noisy sequence to a valid code word of minimal Hamming distance away. There are only two valid code words in the repetition code of length 5, which are 11111 and 00000. All the possible received vectors decoded to 11111 should be closer to 11111 than to 00000. They are binary vectors of length 5 with Hamming weight 5, 4 and 3. There are 16 such binary sequences of length 5.

b) If this code is used in a BSC channel and the crossover probability is p , find the probability P_e of a noisy code word is decoded incorrectly?

A noisy code word is decoded incorrectly only when the Hamming weight of an error pattern is at least 3. Such an error pattern occurs in the BSC channel with probability

$$P_e = \sum_{i=3}^5 \binom{5}{i} p^i (1-p)^{5-i}$$

c) When $p = 0.05$, what is the P_e ?

$$\text{When } p = 0.05, P_e = 10 \times 0.05^3 \times 0.95^2 + 5 \times 0.05^4 \times 0.95 + 0.05^5 \approx 1.15 \times 10^{-3}.$$

4. Prove that it is not possible to find 32 binary code words, each of length 8, such that each code word differs from every other code word in at least 3 places.

It is the same to prove that there is no binary code of length 8 with minimum Hamming distance 3 and 32 code words.

Since $t = \left\lfloor \frac{3-1}{2} \right\rfloor = 1$, $\frac{2^8}{\sum_{i=0}^t \binom{8}{i}} = \frac{256}{1+8} < 32$. This violates the Hamming bound. Therefore, such code does not exist.

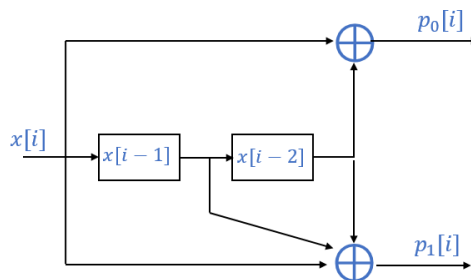
5. A binary (2,1,2) convolutional code with generator $g_0 = (101)$ and $g_1 = (1 \ 1 \ 1)$.

a) Write down the parity check equations.

$$p_0[i] = x[i] + x[i-2]$$

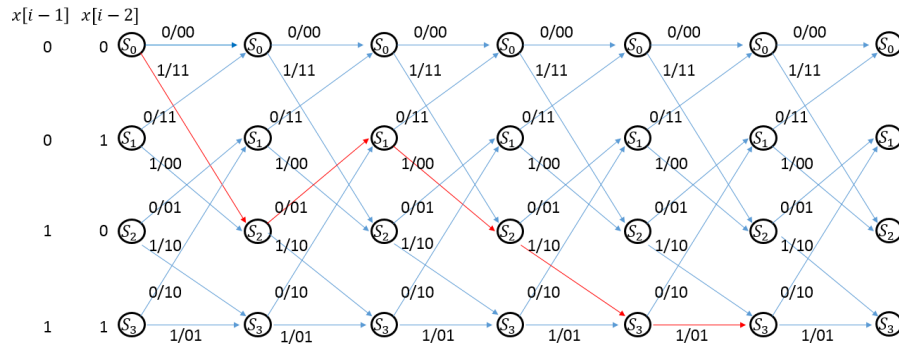
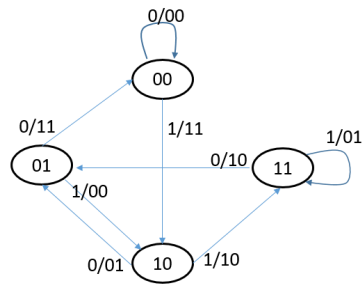
$$p_1[i] = x[i] + x[i-1] + x[i-2]$$

b) Sketch the encoder block diagram and encode the message (1 0 1 1 1).



The message 10111 will be encoded as 11 01 00 10 01 10 11.

c) Find its state machine and trellis diagram. Encode the message (1 0 1 1 1).

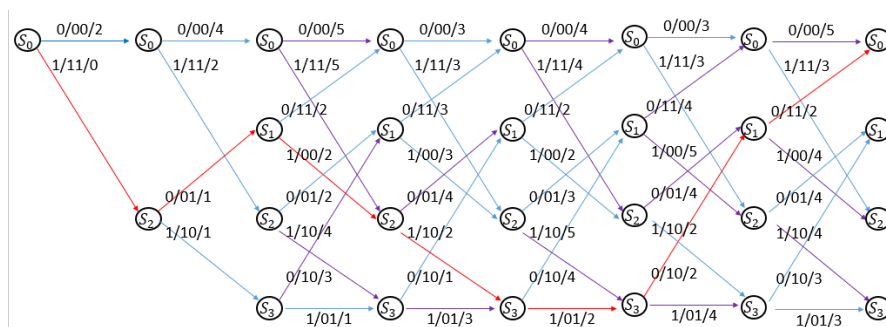


The encoded code word is 11 01 00 10 01 or 11 01 00 10 01 10 11 if two zeros are padded at the end.

d) Find the free distance of this code.

From the trellis diagram, the free distance is 5.

e) Use Viterbi decoder to find the decoding output of received sequence (11 11 01 10 01 10 11).



The output is 1101001001011011.